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Stability evaluation of laser-MAG hybrid welding process

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Outline

[Highlights](#)[Abstract](#)[Keywords](#)[1. Introduction](#)[2. Experiments](#)[3. Results and discussion](#)[4. Conclusions](#)[Acknowledgements](#)

Highlights

- The characteristics of droplets in hybrid arc space are analyzed using high speed camera.
- It studied the effects of laser power to arc power ratio on the weld morphology, droplet transfer characteristics.
- The critical value of the droplet transfer mode transition is determined.

Abstract

This paper describes the laser-MAG hybrid welding of high-strength steel (HSS) formulated with 1.2 mm diameter austenitic stainless steel (ASS) filler wire. A high-speed camera and Hannover analyzers were used to obtain real-time images of droplets and electrical signals during the hybrid welding process. Image processing and mathematical statistics were performed to study how the ratio of laser power to arc power affected the weld morphology, droplet transfer characteristics, instantaneous current probability density. The welding stability was evaluated by using the variation rate of weld width as the main reference. This study provides the fundamentals necessary to select reasonable parameters to obtain better weld stability during the laser-MAG hybrid welding process. The results of this study show that both the laser power and arc power strongly influence both the penetration depth and the width of the weld during the laser-MAG hybrid welding process, respectively. The droplet transfer mode and the ratio of laser energy to arc energy are strongly affected by the distance between the laser and the arc in the laser-MAG hybrid welding process. The droplets have a negligible effect on the keyholes when the distance between the laser and arc is in the range of 2.5–3.5 mm. The welding current is 160 A, which is the critical current of short-circuiting transfer to globular transfer. The critical transition current of globular transfer and projected transfer was shown to occur at a welding current of 180 A. When the welding current is 206 A, the critical transition current of projected transfer and spray transfer is observed. When the welding current is 206–240 A, the fluctuation of the width of weld beam is not significant. When the droplet transfer mode is projected transfer, the welding stability is the best.

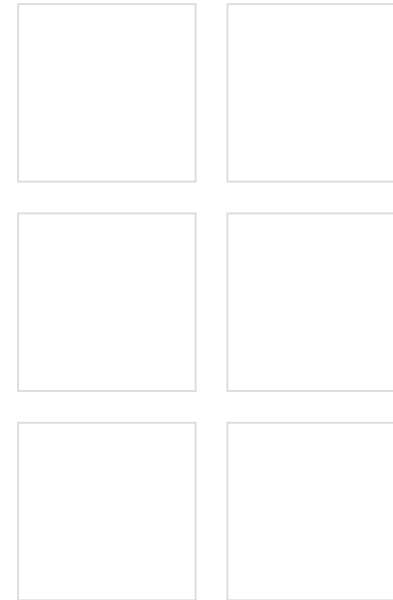
Keywords

High strength steel; Laser-MAG hybrid welding; Droplet transfer; Welding stability

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Figures (17)



[Show 11 more figures](#) ▼

Tables (3)

- ▣ Table 1. Chemical compositions of base ...
- ▣ Table 2. Welding parameters.
- ▣ Table 3. Welding parameters.

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